

SurgeShield / HC-05

District Health Surge Forecast and Emergency Reserve Allocation System

Competition Track: HC-05 | Verification Status: 85/85 Tests Passed | Primary Policy: Fairness Aware

Core Competition Paradigm: Forecast → Quantify Risk → Allocate Reserve → Observe Actuals → Detect Residual Surge → Adapt.

Official Results: Primary Development Seed (20260911) Score = **81.75 / 100** | 20-Seed Robustness Mean Score = **84.02 ± 1.96** | Worst-District Service Ratio = **0.6904** (+11.4% gain over Equal Split). Strictly zero future-data leakage enforced.

1. Problem Statement

During regional public health emergencies and seasonal surges, medical demand across health districts can rapidly exceed fixed local capacities. Health authorities maintain a centralized emergency reserve of **60 capacity units** per month. The operational challenge is to allocate these 60 scarce, non-negative integer units across **12 heterogeneous health districts** under demand uncertainty and unexpected local shocks. The competition scoring awards **30 points** to Forecast Utility (U_{forecast}), **45 points** to Demand-Service Utility (U_{service}), **15 points** to Worst-District Service (U_{worst}), **5 points** to Constraint Compliance, and **5 points** to Runtime/Reproducibility (100 points total). The core imperative is not simply maximizing forecast accuracy, but placing scarce capacity where it prevents catastrophic service failure in the most vulnerable districts.

2. Dataset & Synthetic Demand Generator

The official synthetic data generator generates time-series for 12 districts (D0 to D11) across 42 months (36 historical months $t=0..35$, and 6 future evaluation months $t=36..41$). For district d at month t , underlying mean demand follows:

$$\mu(d,t) = b_d + a_d \cdot \sin(2\pi t/12 + d\pi/6) + g_d \cdot t$$

where baseline $b_d \sim \text{Integer}[60, 160]$, seasonal amplitude $a_d \sim \text{Uniform}[10, 40]$, and linear growth $g_d \sim \text{Uniform}[-0.5, 1.5]$. Noise $\epsilon(d,t) \sim N(0, 8^2)$ produces realized demand $y(d,t) = \max(0, \text{round}(\mu(d,t) + \epsilon(d,t)))$. Nominal district capacity is fixed at $c_d = \text{round}(0.90 \cdot b_d)$, creating structural baseline deficits that require intelligent reserve distribution. In the development scenario (seed 20260911), a hidden surge of +35 units hits D2 and D9 during months 38, 39, and 40.

3. Forecasting Subsystem (Harmonic Regression)

The production forecasting model is an Ordinary Least Squares (OLS) **Harmonic Regression** model with seasonal sine and cosine terms plus a linear trend: $y_{\text{hat}} = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot \sin(2\pi t/12) + \beta_3 \cdot \cos(2\pi t/12)$. Parameter recovery matches the true generator harmonics in closed form without iterative optimization. On rolling-origin leakage-free cross-validation across $t=24..35$, it achieves **$U_{\text{forecast}} = 0.9400$** (MAE = 7.82, RMSE = 9.55), outperforming Seasonal Naive (0.9168), Moving Average (0.9126), and Exponential Smoothing (0.9069). Outputs provide point forecast y_{hat_d} , calibrated prediction intervals, and residual standard error σ_d .

4. Uncertainty & Predictive Risk Estimation

Rather than naively treating point forecasts as ground truth, SurgeShield models future demand as a predictive distribution $Y_d \sim N(y_{\text{hat}_d}, \sigma_d^2)$. Effective capacity after allocating x_d reserve units is $k_d = c_d + x_d$. With standardized variable $z_d = (k_d - y_{\text{hat}_d}) / \sigma_d$, expected unmet demand is computed via the exact analytical **Normal Loss Function**:

$$E[u_d(x_d)] = E[\max(0, Y_d - k_d)] = \sigma_d \cdot [\phi(z_d) - z_d \cdot (1 - \Phi(z_d))]$$

where ϕ is the standard normal PDF and Φ is the CDF. This formula is continuous, strictly convex, and verified against high-precision numerical Gaussian integration to 10^{-14} precision. The discrete marginal benefit of the $(x+1)$ -th reserve unit is:

$$MB_d(x) = E[u_d(x)] - E[u_d(x+1)] \approx P(Y_d > c_d + x).$$

5. Emergency Reserve Allocation Optimization

The allocation optimization problem distributes $R = 60$ integer units across the 12 districts under hard competition constraints: $\mathbf{x}_d \in \mathbb{Z}_{\geq 0}$ and $\sum \mathbf{x}_d \leq 60$. Fractional medical units are never produced. Because expected unmet demand $E[u_d(x_d)]$ is strictly convex and separable, the discrete greedy marginal-benefit knapsack algorithm finds the exact global integer optimum for pure shortage reduction. This was mathematically proven and verified against a Mixed-Integer Linear Program (MILP) using incremental choice variables $\delta_{\{d,k\}} \in \{0, 1\}$ with total unimodularity, yielding bitwise identical solutions.

6. Forecast-Based Fairness-Aware Protection

A pure efficiency knapsack allocator allocates units primarily to large districts with large absolute deficits, starving smaller districts and causing their service ratios ($r_d = 1 - u_d / \max(y_d, 1)$) to collapse, destroying the 15-point U_{worst} score. SurgeShield resolves this via a **Two-Stage Fairness-Weighted Knapsack**:

- **Stage 1 (Fairness Floor)**: Evaluates projected baseline service ratios $1 - E[u_d(0)]/y_{\text{hat}_d}$. Districts below a safety threshold (0.85) are allocated floor units to prevent localized collapse.
- **Stage 2 (Weighted Marginal Efficiency)**: Remaining reserve units are assigned iteratively to: $\mathbf{d}^* = \text{argmax}_d [\mathbf{w}_d(\mathbf{x}_d) \cdot \mathbf{MB}_d(\mathbf{x}_d) \cdot (1 + 0.8 \cdot \text{surge_signal}_d)]$, where vulnerability weight $w_d(x_d) = 1 + \gamma \cdot [\max(0, (y_{\text{hat}_d} - c_d - x_d)/y_{\text{hat}_d})]^{1.5}$ aggressively defends districts with low projected coverage.

7. Online Residual-Driven Surge Detection

The system possesses **zero prior knowledge** of hidden surge locations, magnitudes, or onset months. Anomaly detection operates online from realized residuals $\mathbf{e}_{\{d,t\}} = \mathbf{y}_{\{d,t\}} - \mathbf{y}_{\text{hat}_{\{d,t\}}}$ and standardized Z-scores $Z_{\{d,t\}} = \mathbf{e}_{\{d,t\}} / \sigma_{\{e,d\}}$. Districts transition across states:

- **Normal ($Z < 1.5$)**: Residual within expected Gaussian variation ($\text{surge_signal} = 0.0$).
- **Elevated ($1.5 \leq Z < 2.5$)**: Persistent positive bias detected; close monitoring ($\text{surge_signal} = 0.35$).
- **Anomaly ($Z \geq 2.5$)**: Statistically improbable spike ($p < 0.01$); pre-allocates surge priority ($\text{surge_signal} = 0.70$).
- **Confirmed Surge (2 consecutive shocks $Z \geq 2.0$ or single $Z \geq 3.5$)**: Critical surge state; full adaptation priority ($\text{surge_signal} = 1.0$).

Surge signals legitimately update the forward allocation weight for the subsequent month $t+1$.

8. Sequential Decision Process & Zero-Leakage Architecture

To guarantee strict compliance with competition rules, the evaluation simulates months 36 to 41 under strict temporal causality:

1. **Available History**: At month t , the model only accesses demand history from months 0 through $t-1$.
2. **Forecast & Risk**: Next month t demand is predicted; shortage risks and prior surge signals are assessed.
3. **Reserve Allocation**: Exactly 60 integer reserve units are allocated and committed.
4. **Actual Demand Revelation**: Actual demand $y_{\{d,t\}}$ is revealed ONLY after allocation is locked.
5. **Residual & Anomaly Update**: Realized unmet demand and residual $\mathbf{e}_{\{d,t\}}$ are computed; the surge detector updates its internal state for month $t+1$.

Automated spy tests verify that tampering with future demand has zero effect on current allocations.

9. Official Competition Metrics & Scorecard

Official scorecard weights: $30 \cdot U_{\text{forecast}} + 45 \cdot U_{\text{service}} + 15 \cdot U_{\text{worst}} + 5 \cdot \text{Compliance} + 5 \cdot \text{Runtime}$.

On the development scenario (seed 20260911 with hidden +35 surge into D2 & D9 at $t=38.40$), SurgeShield achieves an official score of **81.75 / 100**, reducing total unmet demand to 2,457 and achieving $U_{\text{worst}} = 0.6573$ in 0.25 seconds.

10. Baseline Policy Comparison & Multi-Seed Robustness

To demonstrate that predictive optimization genuinely outperforms naive strategies, all 5 candidate policies were evaluated through the identical sequential evaluator across **20 random seeds (20260911 to 20260930)** (120 monthly decision cycles per policy):

Policy	U_forecast	U_service	U_worst	Total Unmet	Total Score	Runtime
Equal Split (60/12=5)	0.9483 ± 0.006	0.7752 ± 0.032	0.5763 ± 0.051	2095.8 ± 361	81.98 ± 1.81	0.36s
Forecast Only	0.9483 ± 0.006	0.7822 ± 0.033	0.6124 ± 0.047	2032.0 ± 377	82.83 ± 1.89	0.38s
Risk Based	0.9483 ± 0.006	0.7825 ± 0.033	0.6342 ± 0.042	2029.8 ± 377	83.17 ± 1.93	0.40s
Surge Adaptive	0.9483 ± 0.006	0.7823 ± 0.033	0.6673 ± 0.040	2031.4 ± 378	83.66 ± 1.99	0.43s
Fairness Aware	0.9483 ± 0.006	0.7825 ± 0.033	0.6904 ± 0.035	2029.8 ± 377	84.02 ± 1.96	0.37s

Key Finding: Fairness Aware achieves a mean worst-district utility of **0.6904**, representing a **+11.41 percentage point improvement** over Equal Split (0.5763) and **+7.80 percentage points** over Forecast Only (0.6124), while simultaneously minimizing aggregate unmet demand.

Development Scenario District-Level Breakdown (Seed 20260911):

District	Capacity	Mean FC	Actual Demand	Allocated	Unmet	Service Ratio	Status
D0	57	102.3	603	104	157	0.7396	Protected
D1	113	183.8	1084	70	336	0.6900	Protected
D2	139	185.3	1172	0	348	0.7031	Surge Detected
D3	104	97.2	632	0	35	0.9446	Stable
D4	87	109.5	658	11	125	0.8100	Protected
D5	128	179.5	1081	27	286	0.7354	Protected
D6	96	120.8	749	5	168	0.7757	Protected
D7	86	86.5	588	0	77	0.8690	Stable
D8	121	114.0	695	0	37	0.9468	Stable
D9	134	202.0	1278	36	438	0.6573	Surge Detected (Worst)
D10	81	105.7	649	1	162	0.7504	Protected
D11	85	152.0	904	106	288	0.6814	Protected

11. Reproducibility & Compliance Verification

Given identical scenario seeds, independent evaluations produce bitwise identical forecast, allocation, and residual vectors. All 85 automated tests pass in under 31 seconds. Full constraint compliance is enforced: all $x_d \in \mathbb{Z}_{\geq 0}$, $\sum(x_d) \leq 60$, zero future leakage, and deterministic PCG64 random state.

12. Limitations

- One-Month Surge Lag:** An online anomaly detector can only observe a surge after month t demand is realized; adaptation legitimately begins at month $t+1$.
- Gaussian Residual Assumption:** Normal loss assumes symmetric errors; heavy-tailed Pareto or extreme Cauchy shocks would require empirical quantiles.
- Spatial Independence:** Districts are evaluated independently; cross-district patient transfers or geographic diffusion are not modeled.